

CELLGUARD · DESIGN DECISIONS REGISTER

CellGuard design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

CellGuard design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Open decisions, all proposed, awaiting Amish.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes (spacers and screws, gap pad, cover pillars and notches, fixing positions, fuse holder in line with B+ and copper link, stud terminals, mounting holes, connector position)	(a) accept; (b) ask for changes	(a)	The whole build plan	CGD-DDR-003, A1
2	Cover material	(a) printed flame-retardant polycarbonate, as drawn; (b) sheet aluminium, cut and bent	None made	Cover making (section 3.5, CGD-DWG-104)	CGD-DDR-001 item 12; CGD-DDR-002
3	Cover colour: clear or opaque	(a) clear UL 94 V-0 polycarbonate, so the status light and board can be seen; (b) opaque, as the engineering media show	(a), if a clear V-0 grade can be printed or bought	Cover material bought	Review note, 2026-09-26 render session, item 1
4	Response to R6, board loss 4.44 W against 4 W	(a) 1.5 mΩ-class switches (about 3.5 W); (b) move the protector fuse off the board and count it with the pack fuse	None made	Switch part number; board layout at TRL 4	CGD-CAL-001 v0.3 section 12; CGD-DDR-002
5	Response to R8, state of charge on 20 Ah packs	(a) restate R8 for 50 Ah and larger packs; (b) a periodic full-charge prompt; (c) a 0.5 mΩ shunt	None made	Shunt part (line 5) if (c)	CGD-CAL-001 v0.3 section 12; CGD-DDR-002
6	Response to R14, mass 0.664 kg against 0.6 kg	(a) a 3 mm plate (0.599 kg; tapped M4 holes then have only 3 mm of thread); (b) relax the target to 0.7 kg (0.65 kg was offered before the parts added for construction)	None made	Plate thickness (section 3.2), spacer and screw lengths	CGD-CAL-001 v0.3 section 10; CGD-DDR-002; CGD-DDR-003
7	Low-cost method to check the state-of-charge estimate	For example a calibrated shunt and an electronic load cycling a small pack	None made	First checks at TRL 4	CGD-DDR-001 item 13
8	First test partner	Solar installer, e-bike repair shop or makerspace	None made	Not part of the TRL 3 build	CGD-DDR-001 item 9
9	Status light on the board	(a) add one three-colour status light to line 15; (b) none	(a)	Board layout at TRL 4	Review note, 2026-09-26 render session, item 4
10	Pack layout in product renders		(a)		

#	Decision needed	Options	Recommendation	Affects in the build	Source
		(a) pack beside the board in renders only, engineering media keep the board above the pack; (b) one layout everywhere		Not part of the build	Review note, 2026-09-26 render session, item 2
11	Cover fixing and power cables shown in the product renders	(a) accept as drawn in the renders, updated to the four pillars of CGD-DDR-003; (b) remove	(a)	Not part of the build	Review note, 2026-09-26 render session, items 3 and 5
12	SwapCell project's agreement to the NMC profile and the SwapCell v0.3 CAN message set	Agreement from the SwapCell project	Decided on the CellGuard side (CGD-DDR-002)	Firmware at TRL 4	CGD-PRC-001 open questions; review note cross-repo actions

■ To confirm when parts are bought

Table 2. Items to confirm.

#	What to confirm	Why it matters	Source
1	A protector fuse rated for 40 A continuous at 60 V DC or more, and its price	R4 depends on it; the \$2.00 budget margin may not cover it	CGD-CAL-001 section 4; BOM line 14
2	The fuse holder: base no larger than 44 x 32 mm, four M4 fixing holes at the drawn positions, M8 terminal studs, and fuse blade faces 12.6 mm above its mounting face	The copper link is flat only if the blade faces and the B+ stud shoulder are at the same height; the plate holes follow the holder	CGD-DDR-003 P5; BOM line 8
3	The fuse and holder mass	The mass result keeps the concept figure of 89 g	CGD-CAL-001 v0.3 section 10
4	The stud terminals: M6 male stud, 13 mm shoulder 6 mm tall, through-hole pins	The cable lugs and the link clamp on the shoulder; the shoulder height sets the link height	CGD-DDR-003 P6; BOM line 9
5	The power switch package height of 2.3 mm (TOLT)	It sets the gap pad squeeze with 3 mm spacers; a different height needs a different spacer	CGD-DDR-003 P2; BOM line 4
6	The gap pad: squeezes from 1.0 to 0.7 mm at a load the board fixings can apply, and insulates to at least 100 V	It is the only insulation between the live switch tops and the plate	CGD-DDR-003 P2; BOM line 17
7	The TVS diode and switch avalanche energy against about 1 J	Short-circuit turn-off energy at a 15 μ s delay	CGD-CAL-001 section 4
8	A printable flame-retardant polycarbonate filament (or clear sheet, if decision 4 is (a))	The cover is drawn for printing	BOM line 13

■ Value engineering

Value-engineering target: USD 140.00 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 138.00 (USD 2.00 under the target).

Main cost drivers (CGD-CAL-001, section 11): the eight charge and discharge MOSFETs (line 4, \$20.00), supporting electronics (line 15, \$16.00), the pack fuse and holder (line 8, \$12.00), the aluminium base plate, the main PCB and the secondary protector with its SCP fuse (lines 1, 2 and 14, \$10.00 each). The secondary protector and the parts added for construction (lines 17 and 18) account for \$14.00 of the estimate.

Savings worth trying:

- Confirm the SCP fuse price first, since the \$2.00 under the target could be used up by it (CGD-DDR-003, A2).
- Leave the status light out of line 15 (open decision 9), and decide the cover material (open decision 2) with price in view; a sheet aluminium cover may change line 13 (\$5.00), but no price is established.

■ Decisions made

Table 3. Decisions made.

Date	Decision	Decided by	Record
2026-09-25	BQ76952-class front end; STM32G0B1-class controller building on the Libre Solar BMS firmware; BQ77216-class secondary protector with protector fuse; 40 A rating; high-side switching; LFP and NMC-capable hardware, LFP firmware first; SwapCell v0.3 CAN message set as a profile	Amish: "i accept all your recommendations, go with them across all repos."	CGD-DDR-001, CGD-DDR-002
2026-09-25	Budget raised from \$120 to \$140	Amish, same instruction	CGD-DDR-002
2026-09-25	Reused Apache-2.0 firmware kept in <code>firmware/third_party/</code> with its notices and listed in <code>LICENSE-SOFTWARE</code>	Amish, same instruction	CGD-DDR-002
2026-09-25	Protection settings (SCD 100 mV at 15 μ s, OCD2 40 mV at 20 ms, OCD1 24 mV at 320 ms, OCC 12 mV); top-side cooled TOLT switches; DPAK-class precharge switch; one-point current calibration at build	Amish, same instruction	CGD-DDR-002
2026-09-30	Design for construction: the changes P1 to P8 that make the board buildable	Made under Amish's 2026-09-30 instruction ("fix the design assumptions to match and be physically feasible"); open for his review (open decision 1)	CGD-DDR-003